F318 Pancake Loadcell

Geometry: Shear web structure with high axial and lateral mechanical stiffness.

The F318 features a shear-web construction engineered to provide exceptional stiffness both axially and laterally, making it suitable for demanding measurement environments. This structural arrangement supports the performance expected from universal pancake load cells, especially in applications requiring compact form factors without compromising mechanical integrity.

A key advantage of the F318 is its balanced response when used in either compression or tension. Each bi-directional unit undergoes calibration in both loading directions, ensuring dependable and repeatable output across its full operating envelope. For capacities up to 20kN, the design incorporates counter-bored and threaded outer mounting points, allowing flexible installation while preserving the low-profile geometry.

This model includes a recessed loading boss as standard. For applications that require a raised boss instead, the F254 series may be a more appropriate option; that range is available in capacities from 25kN to 200kN and aligns with many of the functional expectations associated with universal pancake load cells.

Custom engineering support is available for projects requiring modified geometry, tailored mounting provisions, or extended operating conditions. Special versions can be produced with full

Features

- Tension / compression / bi-directional calibration
- Low profile with recessed centre boss
- Fatigue rated on ranges 10kN and above
- Sealed to IP65
- Hardened stainless steel construction
- Traceable calibration with certificate included in the standard price

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compensation for temperatures reaching +250°C, and our engineering team can advise on how best to configure the sensor for your application. Additional information on other load cell types and families is available in the Load cell Specifier Guide.

Overall, the F318 provides a rugged, low-height measurement solution suitable for users who need reliable performance in both compression and tension within a compact shear-web design.

FAQ’s

Can the F318 accurately measure both compression and tension?

Yes. The F318 is calibrated bi-directionally, providing balanced and reliable output in both compression and tension. Each unit is individually verified to ensure repeatable measurements across its full capacity range, maintaining precision in demanding applications.
Can Novatech customise the F318 for special mounting or geometry requirements?

Absolutely. We can modify the F318 with tailored mounting options, altered geometry, or specific loading interfaces. Our engineering team evaluates each request to ensure feasibility while preserving measurement accuracy and the structural integrity of the pancake load cell.
Is the F318 suitable for high-temperature environments?

Yes. Custom versions of the F318 can be manufactured with full compensation for temperatures up to +250°C. This ensures consistent performance and reliable readings even in elevated temperature applications, making it suitable for demanding industrial or laboratory settings.

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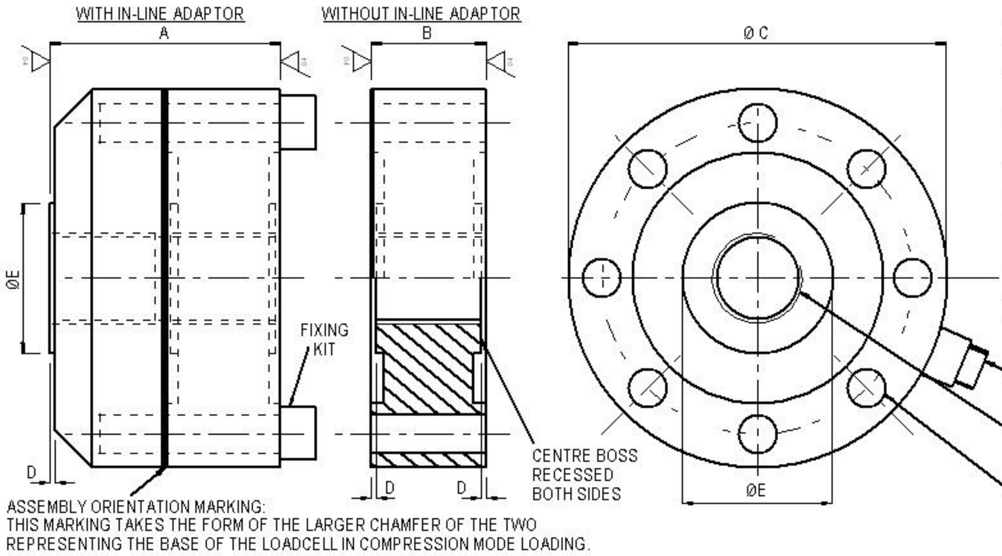
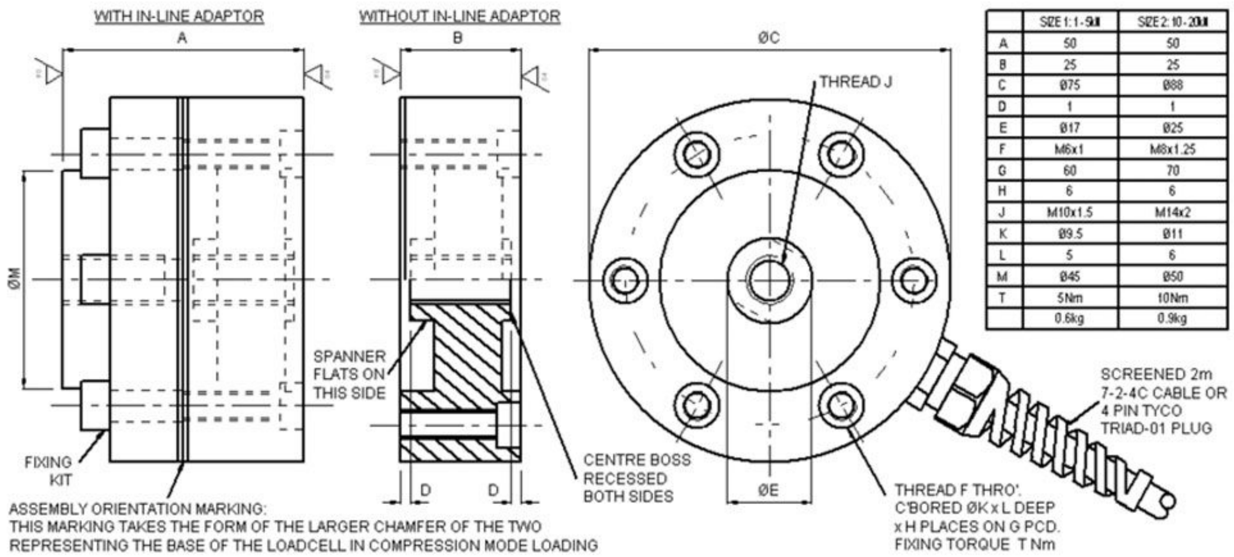
Outline



	SIZE 3: 50 - 200kN	SIZE 4: 500kN
A	92	128
B	46	64
C	Ø151	Ø227
D	2	2
E	Ø60	Ø106
F	Ø15	Ø17
G	124	197
H	8	12
J	M36x3	M64x2
T	110Nm	200Nm
	4.4kg	14.1kg



	SIZE 1: 1 - 5kN	SIZE 2: 10 - 20kN
A	50	50
B	25	25
C	Ø75	Ø88
D	1	1
E	Ø17	Ø25
F	M6x1	M8x1.25
G	60	70
H	6	6
J	M10x1.5	M14x2
K	Ø9.5	Ø11
L	5	6
M	Ø45	Ø50
T	5Nm	10Nm
	0.6kg	0.9kg

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Specifications

Parameter	Value	Unit
Non-linearity - Terminal	±0.1	% RL
Hysteresis	±0.1	% RL
Creep - 20 minutes	±0.05	% AL
Repeatability	±0.02	% RL
Rated output - Nominal	2.1	mV/V
Rated output - Rationalised	2	mV/V
Rationalisation tolerance (applies to single direction calibrations)	±0.1	% RL
Output symmetry	±0.5	% AO
Fatigue life (ranges 10kN and above)	10^8	RL cycles
Zero load output	±4	% RL
Temperature effect on rated output per °C	±0.005	% AL
Temperature effect on zero load output per °C	±0.005	% RL
Temperature range - Compensated	-10 to +50	°C
Temperature range - Safe	-10 to +80	°C
Excitation voltage - Recommended	10	V
Excitation voltage - Maximum	20	V
Bridge resistance	700	Ω
Insulation resistance - Minimum at 50Vdc	500	MΩ
Fixing bolt torque	See outline drawings.	Nm
Overload - Safe	50	% RL
Overload - Ultimate	200	% RL
Sealing	IP65	

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Order Codes

Code	Description
F318CFR0K0	Compression, IP65, unrationalised
F318TFR0K0	Tension, IP65, unrationalised
F318UFR0K0	Bi-directional, IP65, unrationalised
F318CFR0KN	Compression, IP65, rationalised
F318TFR0KN	Tension, IP65, rationalised
F318UFR0KN	Bi-directional, IP65, rationalised
Change the F to a P for the connector version.	

Structural Stiffness

Range (kN)	Stiffness (N/m)
1	1.6×10^{7}
2	1.1×10^{8}
5	3.8×10^{8}
10	1.6×10^{9}
20	1.3×10^{9}
50	1.9×10^{9}
100	4.5×10^{9}
200	9.5×10^{9}
500	1.3×10^{10}

Notes

- AL = Applied load.
- RL = Rated load.
- AO = Average of tension and compression outputs for full load.
- Temperature coefficients apply over the compensated range.

Connections

The loadcell is fitted with 2 metres of PVC insulated 4 core screened cable type 7-2-4C for ranges up to 20kN. Ranges above 20kN are fitted with 16-2-4C cable. A 4 pin Binder chassis plug option is available instead of the cable. For the ranges up to 20kN the plugs are 712 series, for 50kN and above 723 series are used.

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Excitation + = Red or pin 1, Excitation – = Blue or pin 2, Signal + = Yellow or pin 3, Signal – = Green or pin 4, Screen = Orange.

Reverse the signal connections to obtain a positive signal in tension mode. The screen is not connected to the loadcell body.

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